

New systems of learning for distance learning institutions? A six-study review of implementing AIDA

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ABSTRACT

This study examines how a large-scale distance learning institution in the UK designed, implemented, and evaluated an artificial intelligence digital assistant (AIDA) using Sharples' (2025) embedded systems approach. Through six iterative Design-Based Research studies over 18 months, involving 498 students and 20 staff, we explored three questions: (1) how perceptions of AIDA change with experience; (2) its impact on engagement and learning; and (3) how GenAI can be responsibly deployed at a system level. Early Studies 1–3 gathered student preferences for AIDA's design and identified benefits of 24/7, context-specific support alongside concerns over ethics, academic integrity, and data privacy. Later Studies 4–6 explored actual hands-on use by student and staff. In a first explorative randomised controlled trial (Study 6), 115 students were able to use AIDA in an OpenLearn Create course, whereby they spent two times longer and visited more pages relative to the 48 students in the control group. Acceptance increased post-use, especially among initially sceptical participants, with 96% expressing interest in AIDA for formal studies. Mapping the six studies against Sharples' embedded systems approach highlighted enabling factors such as senior leadership sponsorship, cross-unit collaboration, iterative refinement, and data-informed decision-making, while also revealing gaps in institutionalising systems thinking. The findings indicate that purpose-built GenAI tools embedded within the learning environment can enhance engagement and learning outcomes in distance education, provided development is participatory, governance is robust, and integration aligns with institutional strategy. This research offers practical guidance for higher education institutions seeking to adopt GenAI ethically, transparently, and at scale.

1. Introduction

The emergence of generative artificial intelligence (GenAI) systems presents both promise and complexity for higher education institutions (Bond et al., 2024; Dawson et al., 2023; Giannakos et al., 2025; Jin et al., 2025), and distance learning institutions in particular (Rienties et al., 2024; Rienties et al., 2025). While these GenAI tools and approaches have rapidly entered learning ecosystems, substantial questions remain around their pedagogical value, institutional integration, ethical use, and long-term impact on what distance learning institutions are for (Sharples, 2025). Indeed, the recent demise of GenAI chatbot KhanMigo does provide a warning, as learners were not actually engaging with the GenAI chatbot, with limited evidence of positive gains for those who did (Mishra, 2026). This might be particularly surprising as KhanMigo

received substantial funding and support from major tech companies, including early access to ChatGPT, and governments.

A recent study by Jin et al. (2025) amongst 40 top universities in six global regions of their GenAI policy documents suggested that most universities have policies in place for academic integrity and ethical use of GenAI. While brick-and-mortar universities could address some of the academic integrity issues in assessment and feedback with GenAI by for example putting students in large-lecture halls for summative assessments, this is philosophically and practicality unfeasible for distance learning institutions like the Open University (OU) who teach 200K + learners across 50+ countries (Coughlan et al., 2025; Mikroyannidis et al., 2024; Rienties & Ramanau, 2025; Rienties et al., 2025). As 70% of “students” at distance learning institutions are already in/at work, critical discussions are needed in terms of essential skills, competence,

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and knowledge that are appropriate to include in a curriculum in two-five years' time (Fidalgo & Thormann, 2024; Frey & Osborne, 2017).

In line with recent recommendations of Sharples (2025) this study will take an embedded systems approach following Bronfenbrenner (1979) Ecological Systems Theory. The way distance learners engage with their provided online learning environment will be directly and indirectly influenced by various layers and systematic interactions with others (e.g., peers, friends, family members, children), online learning materials, their online class, the broader student community, student services, tutors, IT support, quality assurance, government regulations, tech companies etc. Therefore, in an AI-infused world where technology is omnipresent for distance learners Sharples (2025) has recently suggested an embedded system approach for distance learning organisations to AI, including build awareness and leadership support, fostering collaborative cultures, using data for decision-making, and a commitment to continuous learning.

Therefore, we aim to explore the last-mile extrapolation of implementing GenAI using Sharples (2025)' embedded system approach by critically reflecting on our lessons-learned from a longitudinal, six-study design-based research (DBR) programme conducted at the OU, and its subsequent embedding in "business as usual" systems and processes. We aim to explore the affordances, limitations, and institutional implications of developing effective strategies to address some of the challenges that distance learning institutions face since the arrival of GenAI. In particular, this paper will focus on one state-of-the-art application of co-designing a human-centred AI digital assistant, referred to as AIDA (Mikroyannidis et al., 2025; Rienties et al., 2024; Rienties et al., 2025; Rienties et al., 2025). Drawing on over 500+ student and staff engagements at various levels within the organisation across multiple empirical phases over a period of two + years of data collection, the main aim of this study is: how GenAI can be responsibly designed, deployed, and evaluated for distance learning institutions at a system level?

2. Embedded system approach for distance learning

With the rapid developments in AI and GenAI in particular several system approaches have been developed to make sense of how institutions can best approach the key affordances and limitations of AI (Bond et al., 2024; Jin et al., 2025; Sharples, 2023, 2025; Yang et al., 2021). For example, several systematic literature reviews on Artificial Intelligence in Education (AIED) by Zawacki-Richter et al. (2019) and more recently Bond et al. (2024) have conceptually classified AIED usage in higher education in four broad categories: profiling and prediction; intelligent tutoring systems; assessment & evaluation; adaptive systems & personalisation. Most of the 66 identified studies by Bond et al. (2024) primarily used AIED for profiling and prediction, but as this study excluded more recent GenAI studies one would expect contemporary studies to make headway in terms of assessment & evaluation as well as adaptive systems & personalisation.

2.1. Diffusion of Innovation Theory

Using the Diffusion of Innovations Theory (DIT) Jin et al. (2025) reviewed the adoption policies of GenAI across 40 universities who were in the Top 10 QS rankings in six global regions in terms of innovation, communication, social system, and time. Given the relative innovation advantage of GenAI tools like ChatGPT and the obvious observability of the innovation's benefits and the widely reported use of GenAI tools like ChatGPT by students (Freeman, 2024; Ravšelj et al., 2025; Rodrigues et al., 2024), Jin et al. (2025) reported that most of the included universities had communicated explicit policies in terms of academic integrity and use of GenAI for assessments.

A third and important element of DIT is the social system in which the innovation is adopted, which according to Jin et al. (2025, p. 3) "refers to the norms, structures, and the community's readiness to adopt

new innovations within a given diffusion context". This is particularly important for distance learning institutions to get right due to the separation of space and place between educators and students (Mittelmeier et al., 2021; Rienties & Ramanau, 2025). Establishing social norms and structures might be more complex in distance learning contexts than at brick-and-mortar universities. Furthermore, there is widespread evidence that uptake of new innovations requires both senior level support and buy-in from "the work floor" (Correa-Peralta et al., 2025; Dawson et al., 2018; Herodotou et al., 2020), whereby establishing social norms and structures might be more complex for distance learning institutions than in brick-and-mortar universities.

The fourth element of time is obviously essential from a system perspective in order to explore how a particular innovation has been embedded within an organisation. As highlighted by several recent studies (Bond et al., 2024; Weidlich et al., 2025) most AIED and GenAI studies in particular (Deng et al., 2025) often are relatively short studies with limited longitudinal data. The long-term impact of using GenAI has received limited empirical attention, but a recent study by Kosmyna et al. (2025) using EEG and behaviour measurements of four months of LLM engagement found that students in the LLM condition performed less well at neural, linguistic and behaviour levels relative to those who were only allowed to use Google or just their own brain.

2.2. Sharples' embedded system approach for distance learning

In this manuscript we suggest to adopt Mike Sharples (2025)'s embedded system approach for GenAI as it was specifically conceptualised for a distance learning context. Sharples has adopted Bronfenbrenner (1979) Ecological Systems Theory in a range of EdTech contexts successfully previously, including mobile learning (e.g., Sharples, 2000; Vavoula et al., 2009) and personal inquiry (Sharples et al., 2015), and more recently GenAI (Sharples, 2023, 2025). In line with Ecological System Theory of Bronfenbrenner (1979) the way distance learners interact with their provided online learning environment and GenAI in particular will be directly and indirectly influenced by various layers and systematic interactions with others (e.g., peers, friends, family members, children), online learning materials, their online class, the broader student community, student services, tutors, IT support, quality assurance, government regulations, tech companies, etc.

While there are similarities between face-to-face institutions and distance learning institutions, distance learning institutions often serve a substantially different student population (Hlosta et al., 2022; Nguyen et al., 2020; Richardson et al., 2020), primarily staff and students work at a distance (Herodotou et al., 2023; Richardson et al., 2015), and often have open entry for undergraduate degrees (Olney et al., 2021). With a national and sometimes international reach distance learning institutions often provide online learning experiences rather than local campus-based experiences (Mittelmeier et al., 2021; Rienties & Ramanau, 2025), thereby potentially exacerbating some of the academic integrity issues. Similarly, recent big data studies on assessment practices suggest that most modules and qualifications at the OU could be passed with GenAI tools irrespective of assessment types and disciplines (Hardie et al., 2024), fundamentally questioning whether (distance learning) institutions should continue to assess, and if so what for (Sharples, 2025). Therefore, in an AI-infused world where technology is omnipresent for distance learners Sharples (2025) has recently suggested an embedded system approach for distance learning organisations as illustrated in Table 1.

2.3. AIDA overview

We specifically chosen to focus on the Artificial Intelligence Digital Assistant (AIDA) out of over a dozen of other AI initiatives at the OU because:

Table 1
Sharples's embedded system approach to AI for distance learning.

Agenda Item	Key Actions	Purpose
1. Build awareness and leadership support	- Educate leaders in systems thinking - Highlight benefits for managing complexity and sustainable innovation	Foster informed and committed leadership to support systemic transformation
2. Develop champions	- Identify influential individuals (e.g., leaders, administrators, teachers) - Empower them to advocate for systems approaches	Create internal advocates to lead and sustain change in complex environments
3. Map the institution as a system	- Use tools like system maps, causal loop diagrams - Understand interconnections across functions - Involve diverse stakeholders	Gain a holistic understanding of the institution's structure and dynamics
4. Define innovation goals	- Set clear, shared objectives for innovation - Align innovation goals with institutional mission and values	Ensure purposeful and mission-aligned innovation initiatives
5. Identify leverage points	- Locate small changes with high potential impact using system maps - Prioritise initiatives by feasibility and impact	Focus resources where they can make the biggest difference
6. Foster collaborative cultures	- Break down silos - Form interdisciplinary teams - Enable open dialogue and stakeholder engagement	Encourage cooperation and trust across departments and stakeholder groups
7. Develop iterative processes	- Pilot small-scale innovations - Use feedback loops for refinement and scaling	Promote experimentation, learning, and adaptability in innovation
8. Build systems thinking capacity	- Provide training and professional development - Partner with external systems-thinking experts	Equip staff and faculty with tools to understand and navigate complexity
9. Use data for decision-making	- Monitor KPIs and systemic health metrics - Analyse trends and feedback loops	Make informed, evidence-based decisions and identify systemic issues early
10. Institutionalise systems thinking	- Embed principles in strategy, policy, and operations - Include systems thinking in curricula for staff and students - Commit to ongoing evaluation and learning	Sustain long-term change and foster a culture of critical reflection and continuous improvement

Adapted from Sharples (2025).

- 1) AIDA contains a number of design features that we believe are important for the future of AI-based learner support – see below.
- 2) AIDA has used a robust Design-Based Research approach of six studies (Rienties et al., 2025; Rienties, Ullmann, et al., 2025);
- 3) AIDA has been co-developed, tested, and evaluated by an interdisciplinary team of three units (Knowledge Media Institute (KMI), Institute of Educational Technology (IET), and Digital Services) (Coughlan et al., 2025; Mikroyannidis et al., 2024; Rienties et al., 2024; Rienties et al., 2025; Rienties, Ullmann, et al., 2025);
- 4) Relative to other GenAI initiatives developed at the OU, AIDA has been tested extensively with students and staff after 18 months, received extensive senior management support, and the results of the various design-iterations have been reviewed and subsequently published in various journals.

- 5) Despite substantial testing, developing and refinement and evidence of support by students and staff, the actual upscaling of AIDA as business as usual has thus far not materialised.

Core to AIDA's design is the recognition that both the accuracy of generated responses and their pedagogical appropriateness are critical if generative AI is to be used responsibly and productively in education. In contrast to generic large language model (LLM) deployments, which are prone to hallucination and decontextualised answers, AIDA was explicitly architected to prioritise provenance, transparency, and pedagogical intent (Mikroyannidis et al., 2024; Rienties, Bektik, et al., Under Review; Rienties et al., 2025). From the outset, we therefore moved beyond a “stand-alone” LLM approach and adopted a staged retrieval-augmented generation (RAG) strategy, subsequently extending this to a GraphRAG architecture, where all responses are grounded explicitly in curated OU content.

In practice, this means that AIDA retrieves relevant learning materials—such as course texts, activities, and guidance—from trusted OU sources and conditions its responses directly on this material (Mikroyannidis et al., 2024; Rienties, Ullmann, et al., 2025). Crucially, for every response provided, AIDA presents direct links back to the originating OU learning resources, enabling students to inspect sources, verify accuracy, and interact critically with the material. This explicit surfacing of provenance was a deliberate design choice to support epistemic trust, reduce hallucination risk, and reinforce good academic practice, rather than positioning AIDA as an unquestionable authority.

To support this architecture at institutional scale, AIDA was developed and deployed within the OU's Azure Enterprise platform, providing a secure, compliant, and extensible foundation for integrating state-of-the-art AI capabilities into core learning systems. This enterprise deployment allowed AIDA to take advantage of newly released large language models as they became available, without requiring architectural changes to the surrounding system. During the studies reported here, the primary model used was GPT-4o, selected for its strong performance across multimodal reasoning, instruction following, and low-latency interaction—capabilities that are particularly important for real-time learner support. The use of an enterprise cloud platform also enabled appropriate governance, monitoring, and data-management practices consistent with institutional requirements for responsible AI deployment.

However, ensuring factual accuracy alone is insufficient for meaningful learning support. A second, equally central design principle underpinning AIDA is the recognition that learning is a structured pedagogical activity (Giannakos et al., 2025; Mikroyannidis et al., 2025; Sharples, 2023, 2025; Weidlich et al., 2025), not merely a sequence of questions and answers. To address this, we encoded key elements of pedagogical knowledge within a dedicated Knowledge Graph, allowing AIDA to reason not only about what content to provide, but how, why, and in what role it should support the learner. This pedagogical knowledge representation captures the learning task that the student is engaged in (e.g., concept formation, revision, assignment preparation), the pedagogical approach to be applied (e.g., constructivist, social, or inquiry-based learning), the role AIDA should adopt in that context (e.g., explanatory guide, Socratic challenger, quiz master, or study mentor), as well as the academic domain, course level, and interaction language. By combining this structured pedagogical context with domain-specific content retrieval, AIDA is able to adapt both the form and function of its interactions, moving from simple informational responses to scaffolding, prompting reflection, formative feedback, or activity generation aligned with established OU pedagogical frameworks. Together, the integration of GraphRAG-based content grounding, a pedagogically explicit knowledge graph, and an enterprise-grade LLM infrastructure enables AIDA to operate as a learning-aware digital assistant rather than a generic conversational agent.

2.4. Research questions

As indicated in Fig. 1, we used a Design-Based Research methodology with three cycles of iterations aiming to co-design, test and iterate AIDA with students, staff and senior management. According to Wang (2020, p. 2) “[t]he whole DBR process is not carried out in a linear sequence, but rather includes recursive and iterative nested processes, and the possible solutions which are the result of the DBR processes include new technological products, educational products, a technological enhanced process, and programs.” In total, six studies were conducted between December 2023 – July 2025 with in total 498 students and 20 staff at The Open University (OU) in the UK. For a detailed description of the design assumptions, design principles of AIDA and how these were adjusted over time in the first five studies we refer to Rienties et al. (Submitted). Study 6 was the first explorative study where volunteering OU students could use and interact with AIDA “in the wild” in May-July 2025, whereby in a randomised control trial setting 163 participants were enrolled in the experimental (with AIDA $n = 115$) or control condition (no AIDA $n = 48$) of two short (non-credit bearing) online courses.

In this six study approach we use Sharples's embedded system approach to explore how institutionally developed GenAI systems like AIDA can be responsibly designed, deployed, and evaluated for distance learning institutions. Building on our overall research aim, we will specifically address the following three research questions:

- 1) What perceptions do participants have of AIDA and do these change once they have experience of using it?
- 2) How does the inclusion of AIDA change engagement with learning, including progress in the course and assessment of learning?
- 3) How GenAI can be responsibly designed, deployed, and evaluated for distance learning institutions at a system level?

3. Methods

3.1. Setting

The OU is Europe's largest institution for distance education, with a student population of approximately 200,000. It offers more than 200 accredited programmes spanning disciplines such as the arts and humanities, business and law, education, health and social care, and various branches of science. Most teaching is delivered online, with students receiving guidance and support from associate lecturers, each of whom usually works with a cohort of about 20 learners. These studies sit within a wider institutional at the OU which we will outline and then map against Sharples embedded system approach. The OU has two units dedicated to research and innovation related to teaching and learning. The Institute of Educational Technology (IET), setup at the founding of the university to explore new pedagogical approaches related to distance teaching and the Knowledge Media Institute (KMi), setup in 1995 to research how new media and knowledge technologies could impact OU teaching. The two units formed the core of the multi-unit collaborative team that carried out the work described in this paper.

Both units have over the years developed technology platforms which have had an impact on OU teaching. For example, iSpot a citizen science platform created in IET has 80,500 registered users who have posted over 1,700,000 images from 180 countries recording more than 43,000 different biological species (Scanlon et al., 2014). *OU Analyse*, developed in 2011 by KMi, is a machine-learning based learning analytics tool which raises pass rates by 7% in students (10% for BAME students) by accurately predicting student outcomes and allowing our tutors to focus their energies to students at risk (Herodotou et al., 2020, 2023; Hlosta et al., 2022).

Digital Services is the third unit involved in upscaling and mainstreaming innovations into business as usual at the OU, who maintain over 250 OU services and systems that students and staff use every day. Digital Services became involved from Study 5 onwards to ensure that lessons-learned from the research and explorative studies with students and staff could be embedded into any mainstream applications in the

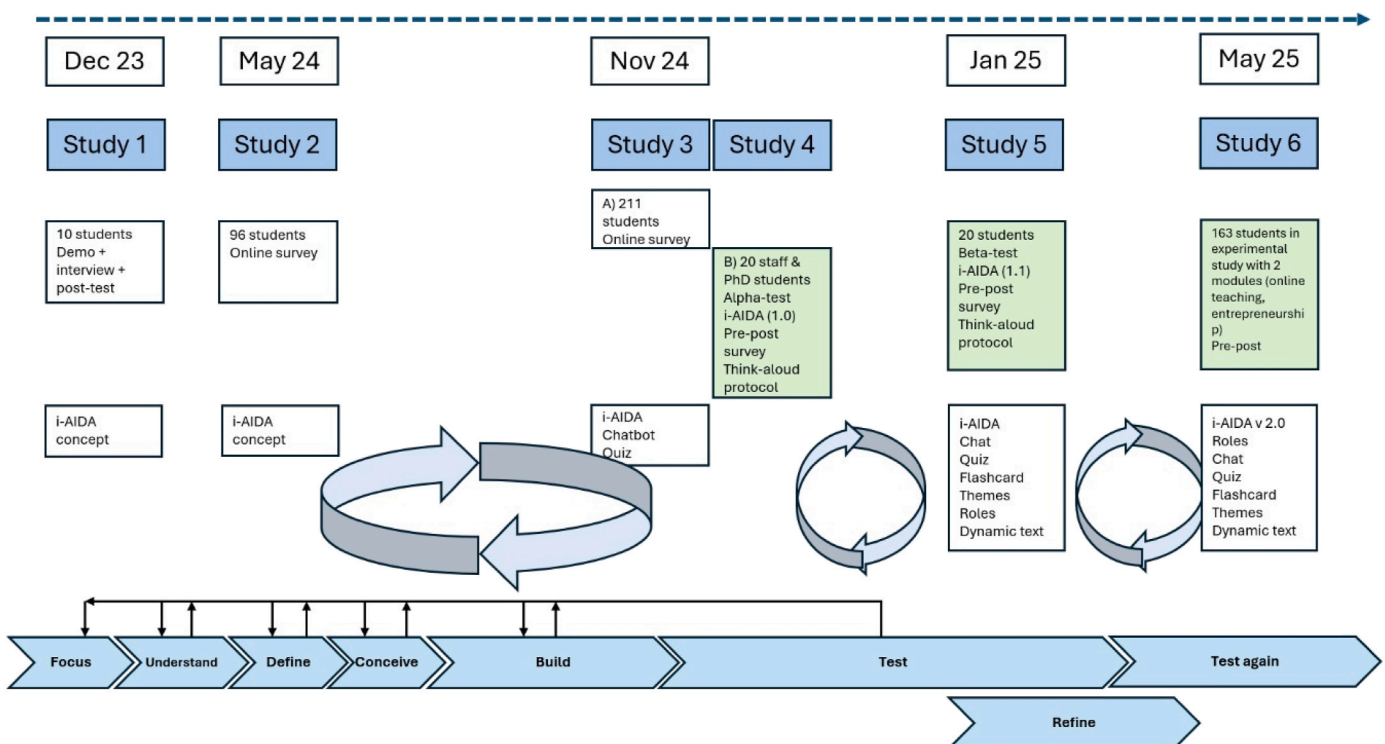


Fig. 1. Six-studies overview of DBR approach of AIDA.

virtual learning environment (VLE) of Moodle.

3.2. Participants

In total 498 students and 20 staff participated in the six Studies. For a detailed description of the participants we refer to the respective Studies and Table 2. As Studies 1-5 have been published we primarily added more details about Study 6 and Sharples' embedded system approach, and the current state of play of AIDA (i.e., April 2026). For Study 6 a randomised control trial approach was employed where a group of 48 participants were randomly assigned to the control condition of their chosen course (i.e., Take your Teaching Online; Employee Engagement) without AIDA. A pragmatic decision was made to integrate AIDA on sandbox version of OpenLearn Create (OLC: an open and free MOOC-like platform run by the OU) rather than the main Moodle VLE. In total 115 participants were assigned a version of their chosen course with AIDA enabled. The two short online OLC courses had around 20 h of study materials available, and participants were invited to explore their registered course and use the functionalities (with or without AIDA), design, and materials of the respective course. After interacting with the OLC course for some time participants were encouraged to complete a post-test to reflect on their experiences.

3.3. Instruments

As indicated in Table 2 a range of methods and approaches (including pre-post-test, interviews, think-aloud, and prompt analysis) were used in the three design-iterations of AIDA, whereby for Study 6 we used a randomised control trial. For a detailed description of the participants, instruments, and specific findings we refer to the respective Studies 1-5. As illustrated in terms of the average age of participants, these distance learners are substantially older than those GenAI studies reported elsewhere (e.g., Deng et al., 2025; Kosmyna et al., 2025; Ravselj et al., 2025).

In terms of Study 6, like Study 5 a pre-post test design was used using the previously validated instruments from Study 1-3. In total 209 participants completed the pre-test which consisted of around 20 questions related to participants' prior GenAI experience, their perspectives on AIDA in general and one developed by the OU in particular, as well as any challenges related to AI. Furthermore, participants could choose from one of two courses (i.e., Take your teaching online, Employee Engagement) and subsequently answered three knowledge questions. Around 20% of participants had no experience with GenAI, but most had some to substantial experience with GenAI. In addition, pre-post knowledge tests were included based upon the preferred course that participants wanted to follow.

3.4. Procedure and data analysis

An interdisciplinary team of 15+ educators, tech specialists, designers and evaluators across three units of KMi, IET, and Digital Services (from Study 5 onwards) designed and evaluated the various versions and iterations of AIDA in the six studies. In total four facilitators supported the six studies of the AIDA tool by participants. For RQ1-2 for Study 6 SPSS 29 was used to analyse the pre-post data as well as the user data. For RQ 3 the four authors of this manuscript, who are all Professors and senior academics in the OU and extensively involved in various GenAI projects, used the Sharples's embedded system approach to AI and independently coded the 10 agenda items per study, as well as their understanding of the broader GenAI context within the OU. Subsequently, following discussions consensus was reached across the four authors on the respective coding. According to Web of Science, the authors have an average of 99 publications ($SD = 55.63$), and an average H-index of 23.75 ($SD = 12.60$). The authors engaged in a highly collaborative and iterative process across the initial submission of this manuscript and the subsequent revisions in April 2026. Over an

intensive two-week revision period, the team of four authors undertook sustained critical dialogue, reflexive interrogation of the findings, and repeated refinement of the interpretations and discussion, thereby strengthening the analytical rigour, coherence, and scholarly contribution of the final manuscript. For a detailed description of the educational/pedagogical data analytics we refer to Table 2 and published studies. This research received Human Ethics Research Approval for the various Studies (e.g., HREC/2024-0660-2, HREC/2025-0831-3).

4. Results

4.1. Perceptions of AIDA by stakeholders

In the first three Studies we specifically wanted to *focus*, *understand*, *define*, and *conceive* what features an AIDA should have according to three cohorts of distance learning students who received various iterations and prototypes and screenshots of potential AIDAs, as illustrated in Fig. 2. Across Studies 1-3 where participants were asked about their perceptions without actually using a "real" AIDA. The most preferred reason for the 315 participants was that AIDA would be available 24/7 and would provide immediate feedback in the academic setting of the learner, rather than providing generic content such as ChatGPT (Rienties et al., 2024; Rienties et al., 2025; Rienties et al., 2025).

Another key factor identified by participants was the value of embedding an AIDA directly within their OU learning environment, rather than hosting it on a separate platform. The potential for tailored, context-specific responses—aligned to individual learners' needs and their progression through course content—was the third most frequently cited benefit. For OU students, whose studies are entirely online, this was considered particularly relevant. Despite generally positive attitudes towards AIDA, participants also voiced notable concerns, especially in relation to academic integrity, ethical and societal implications, and the protection of personal data. Even so, roughly 60% of students across the three Studies indicated they would like to use an AIDA to support their learning.

A first workable alpha version of AIDA (Study 4) was tested with 20 staff in Edtech/AI in November/December 2024; Coughlan et al., 2025), after which a beta-version of AIDA (Study 5) was developed and subsequently tested with 18 students in January 2025 under the supervision of an online facilitator (Rienties, Ullmann, et al., 2025). In total nine students at the pre-test indicated that they would want an AIDA for their studies, four were neutral, and five disagreed. As indicated by Rienties, Ullmann, et al. (2025) the 18 participants were much more positive about AIDA after engaging with AIDA for around 45 min, which was significant with a large effect size ($t = 3.482$, $p < .01$, $d = .81$). Participants' perspectives substantially improved after engaging with AIDA, in particular for those who were initially sceptical.

Participant feedback on AIDA's interface and functionality revealed a mix of positive impressions and suggestions for refinement. The quiz tool was particularly well received, with students praising the immediacy of feedback and the detailed explanations provided for incorrect answers. Several respondents recommended adding options to adapt quiz content to their specific learning objectives. Engagement and motivation emerged as the second most frequently mentioned theme (14% of comments), with remarks largely favourable. Opinions on content delivery and overall user experience were more varied: while many appreciated the clear button layout and informative drop-down menus, others pointed to ways the system could be made more user-friendly and accessible. Overall, the interface was considered suitable for its stage of development, though participants stressed the importance of further enhancing usability and accessibility to ensure that AIDA remains inclusive and engaging for a wide range of learners (Rienties, Ullmann, et al., 2025). These design suggestions were included in the final design iterations for Study 6.

Table 2
Overview of six empirical Studies.

	Study 1	Study 2	Study 3	Study 4	Study 5	Study 6
Participants	10 students	96 students	211 students	20 OU Staff	18 students	163 students
Recruitment	A random sample of 400 students studying at a range of levels and disciplines were extracted from institutional database. Two reminder emails were sent for volunteers to participate in 1 h online interview.	Same recruitment approach but sample of 2000 students.50% were in year 1, and rest in year 2-Post-Graduate. Two reminder emails were sent for volunteers to participate in 10-15 min online survey.	Same approach as in Study 2 with 2000 students	60+ colleagues and PhD students from 2 R&D units approached via email to participate in 1 h alpha-test of i-AIDA	85 participants from Study 2-3 were willing to join follow-up interview. In total 27 students expressed an interest to join, and we stopped the recruitment process at 20 participants	2 * 2000 email invites were sent as in Study 2
Response rate	3%	5%	11%	30%	50%	4%
Average duration	Interview: 54 min (range 45:17 to 1:16:58). Survey: 12:27 (range 2:58 to 28:55)	Survey: 18:05 (range 2:44 to 1:31:12)	Survey: 15:26 (range 2:12 to 1:19:41)	Alpha-test: 43:56 (range 29:34 to 56:51)	Beta-test: 44:16 (range 28:08 to 55:47)	Time spent Open Learn: 62m42 (SD = 354.20)
Instrument	Online Demo & interview Follow-up online survey 18 Likert-response questions 5 open questions	Online survey 25 Likert-response questions 15 check-box items for Gen AI usage 5 open questions	Same survey as in Study 2 with two screenshots of AIDA	Alpha-test Pre-post survey Think-aloud protocol	Beta-test Pre-post survey Think-aloud protocol 6 Likert-response 2 Open questions Comparison on AI Digital Assistant Demographic questions	Randomised Control trail Pre-post survey including knowledge measurement Objective measurement of engagement with AIDA Time and pages visited in OLC
Educational/ pedagogical data analytics	Quantitative analysis in SPSS 27 with Cronbach A of .686-.702 Thematic analysis of two themes.	Quantitative Analysis in SPSS 29 using ANOVA. Thematic analysis of 8 themes with Cohen K .659-1 across 3 coders.	Quantitative Analysis in SPSS 29 using ANOVA, k-means cluster analysis, correlation. Thematic analysis across three clusters of students with Cohen K .659-1 across 6 coders in total.	Quantitative Analysis in SPSS 29 using paired sample <i>t</i> -test Think-aloud screencasts, on-screen navigation and concurrent verbalisation while completing the predefined tasks, and the facilitator logbooks documented for each task, whether it was completed, the key narratives expressed during use, design suggestions, and any bugs or usability issues encountered and facilitator logbooks Thematic analysis of 10 themes.	Quantitative Analysis in SPSS 29 using paired sample <i>t</i> -test Think-aloud screencasts, on-screen navigation and concurrent verbalisation while completing the predefined tasks Conversational data and prompt analysis of participants Facilitator logbooks documented for each task, whether it was completed, the key narratives expressed during use, design suggestions, and any bugs or usability issues encountered and facilitator logbooks	Quantitative Analysis in SPSS 29 using pre-post attitudes and knowledge, objective engagement data in OLC, correlations, Thematic analysis across 10 agenda items of Sharples (2025) with 4 coders.
Time of measurement	December 2023- February 2024	May – June 2024	November 2024	Nov – Dec 2024	Jan 25 – Feb 2025	June-July 2025
Female	60%	52%	63%	65%	66%	54%
Age	40.00 (SD = 13.09)	52.30 (SD = 16.61)	45.84 (SD = 16.37)	n/a	50.05 (SD = 14.65)	55.00 (SD = 20.53)
Data previously reported in	Rienties et al. (2024)	Rienties, Tessarolo, Coughlan, Herodotou, et al. (2025)	Rienties, Tessarolo, Coughlan, Coughlan, et al. (2025)	Coughlan et al. (2025)	Rienties, Tessarolo, Coughlan, Coughlan, et al. (2025)	New data

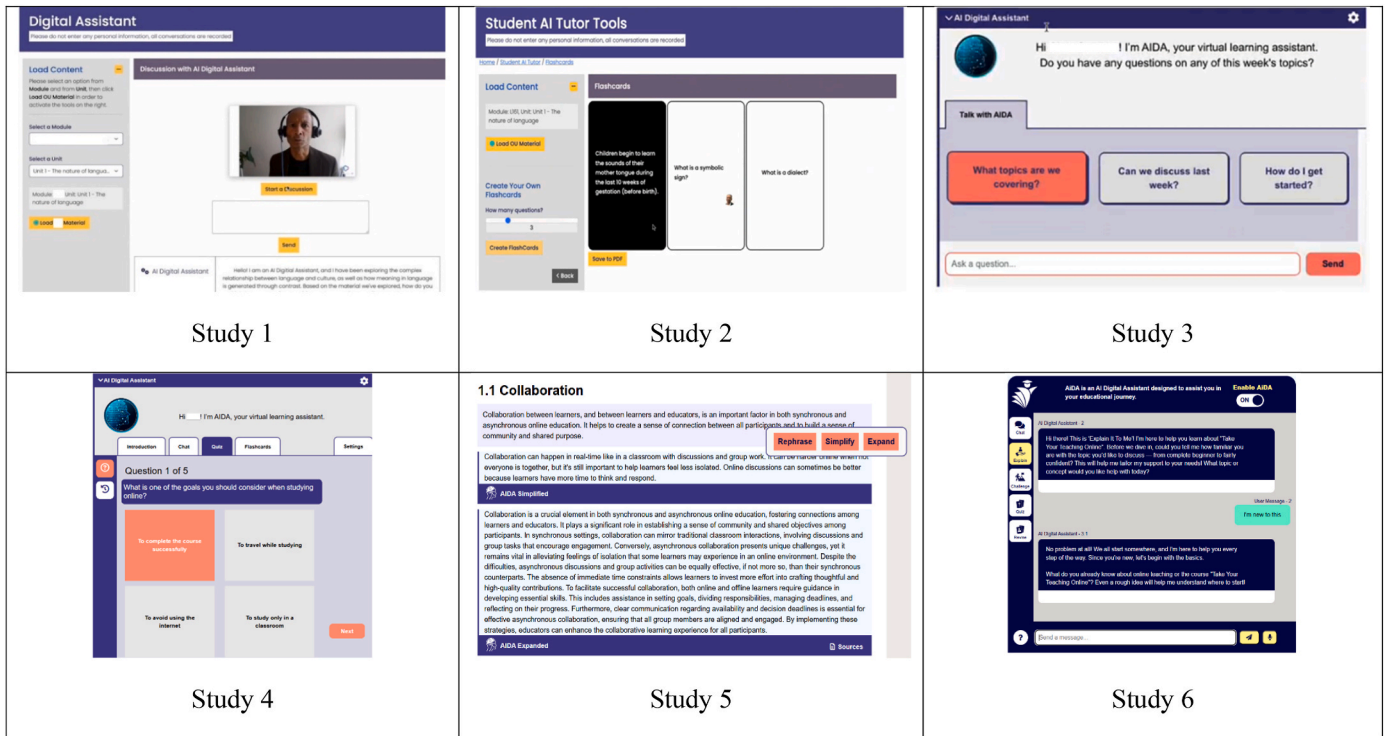


Fig. 2. Design iterations of AIDA of Six Studies.

4.2. RQ2 influence of AIDA on usage, progression and knowledge

Study 6 was the first explorative study where OU students could explore and use AIDA. At the pre-test 82% of 209 participants were positive about having an AIDA in general for their studies, while 89% of participants thought it would be useful to have an AIDA specifically developed by the OU. Nonetheless 62% of participants indicated that there were challenges with AIDA (e.g., ethics, integrity). Broadly these results are similar to previously reported findings from Studies 1-5, and the Cronbach alphas of the respective scales suggested similar and reliable scales, and suggesting that the participants in Study 6 were broadly similar to their peers in previous studies. As indicated in the correlation matrix of Table 3, those who had relatively more experience with GenAI were more positive about the affordances of AIDA and AIDA for the OU in particular, which is to be expected.

In total 163 (78%) out of 209 participants who completed the pre-test subsequently enrolled in OLC for one of the two courses. No significant differences in the pre-test scores were found between those who enrolled in OLC and those who did not, and also no differences were found in terms of who were enrolled in the control or experimental condition, suggesting no imbalance in student characteristics.

The 48 participants who were enrolled in the control condition without support of AIDA spent on average 30.85 min (SD = 59m33) using in OLC. In contrast, the 115 participants who were supported with AIDA spent on average twice longer in OLC (M = 75m59, SD = 419m40), whereby in particular the large standard deviation is noteworthy. Using Mann-Whitney *U* test, no significant differences were found ($Z = -1.433$, $p = ns$). This indicated that many students just briefly used in OLC (in line with the brief to just explore their chosen OLC course), irrespective of the condition.

50 students (17 in control condition, 33 in experimental condition) completed the post-test. In terms of assessment of learning, participants completed a short three item self-assessment of their respective knowledge and expertise at the beginning and end of the OLC course. The Cronbach Alphas suggested a good and consistent reliability of the knowledge assessments. When combining the knowledge tests across the two courses a higher learning gain in the experimental AIDA group ($n = 33$) relative to the control group ($n = 17$) was present ($F = 4.796$, $p < .05$). However, given the relatively small sample of volunteers who completed the post-test and the imbalance between the two conditions we would caution about these preliminary results.

96% of students were keen to have an AIDA available in their studies.

Table 3
Pre-test and correlations of Study 6.

	#items	Mean	SD	Cronbach alpha	1	2	3	4	5	6	7	8
1. Gen AI experience (0-1)	10	.27	.23	.756								
2. AIDA expected services (1-5)	4	3.95	.76	.705	.321**							
3. AIDA specifically for OU	5	4.21	.70	.865	.334**	.645**						
4. AI challenges	5	3.65	.83	.789	-.132	-.102	-.065					
5. Pretest Employee Engagement (1-5)	3	4.25	.62	.809	.330**	.413**	.403**	-.027				
6. Pretest Take your teaching online (1-5)	3	3.67	.72	.612	.235*	.112	.219*	.005				
7. Pretest combined (1-5)	3	3.99	.73		.309**	.220**	.275**	.022				
8. Course (1-2)	1	1.22	.78		.106	-.019	.141*	.178**	.117	.062	.275**	
9. Included in the experimental AIDA condition (0-1)	1	.71	.46		-.089	-.045	-.066	-.072	-.068	-.021	-.071	-.033

N = 209, * $p < .05$, ** $p < .01$.

The control group did not have actual experiences with AIDA, but at the post-test they were given screenshots of AIDA like previously done in Study 2-3 and were asked whether or not these would be useful for them. In terms of participants' perceptions of the AIDA functionalities, those in the experimental group were most positive about all the core AIDA functionalities, with average Likert Response ratings well above 4 (out of 5). The most liked function was the Quiz function, that provides 5 multiple-choice questions per visited page, followed by the Explain chat function, and the "normal" chat function as indicated in Table 4. The differences between these six functions were relatively small.

4.3. RQ 3: Responsible design and evaluation of GenAI for distance learning

Finally, in terms of Sharples (2025) embedded system approach for GenAI we have unpacked the various stakeholders and layers of interactions with others in Table 5. As is indicated in Fig. 1 and Table 5 these six Studies did not emerge in isolation, and were co-developed and co-designed bottom-up with the main stakeholders of students and staff, while at the same time with strong senior institutional commitment.

In terms of building awareness and leadership support, in a large organisation like the OU it is essential to get commitment from both senior management and from the work-floor. Early pilots and experiments on the use of AI were funded by a dedicated innovation activity (Test and Learn) for a number of years in partnership with a large IT provider. This allowed us to conduct quick experiments – of around 6 weeks to 3 months – to guide longer work. One such pilot was around AIDA, using Generative AI (GPT-3) in January 2023. The AIDA work reported here was funded under the continuation of this innovation activity named the 'Innovation Foundry', and supervised by a steering board chaired by the OU's Pro Vice-Chancellor for Research and Innovation (PVC R&I). After the first two explorative Study 1-2 the results were presented to the two Pro-Vice Chancellors (PVC R&I and PVC Students) which helped to gain both organisational and financial support.

The findings from Study 3 and Study 4 were subsequently shared in Quality Enhancement Reports (QEI) with PVCs, and several demos of AIDA following Study 5 were provided at various platforms, including PVCs. The third iteration of AIDA and the results of Study 6 were presented at Vice Chancellor Executive (i.e., the main executive team) and Council (the OU's executive governing body) in June-July 2025. The strong connection of the work to PVC R&I led to dedicated presentations at a number of Executive Away Days and to OU Council (responsible for strategic planning and monitoring effectiveness and performance). Running in parallel, an AI Steering Group sponsored by three executive members in April 2025, which two of the paper authors sits on. Led by the OU's Chief Data Officer and chaired by a new Director of AI Office. This group is sponsored at executive level by the PVC Students and the Chief Digital and Information Officer. Subsequently, a specific AIDA Steering Board was set-up in August 2024 following Study 1-2.

Various studies have shown that developing champions within an organisation can help to embrace change (Herodotou et al., 2020;

Vigentini et al., 2020). In the first four Studies our focus was on getting to a reasonable AIDA that meets the needs of our distance learners, but since January 2025 we proactively demoed AIDA at various staff meetings and specifically engaged several Faculties like Faculty of Business and Law (which was one of the two courses in Study 6). Similarly in the summer of 2025 we worked with another Faculty (WELS) to help with an optional induction course for students supported by an updated version of AIDA, which was made available to 354 students who were new to the university in September-October 2025.

In terms of mapping the institution as a system a lot of work over the last 20 years has been done mapping the various processes as the OU works at scale. With the arrival of GenAI the AI steering group is reviewing whether these mapping need to be changed. Within the respective Studies the engrained knowledge of the interdisciplinary team of the organisation was used, but no formal mapping was done.

The defining of innovation goals was a specific purpose of AIDA and were explicit from Study 2 onwards. Four main services were identified in Study 2 and from Study 3 specific design specifications and principles for AIDA were defined. On an institutional level, the AI Steering Group has a process for receiving suggestions and prioritising/advising on these, but most of the "heavy-lifting" of the initial AIDA was done by KMi and IET.

The fifth element of identifying leverage points was to involve Digital Services and OLC as a platform to launch AIDA into the student experience from Study 4 onwards. As the OU works at scale and implementing innovations in the live production environment requires substantial investment, time, and testing, we opted to use the OLC platform for quick testing and evaluation. At the same time, on an institutional level the OU has been conducting quick pilots with AI to ascertain potential wins, including accessibility (Coughlan & Iniesto, 2025) and predictive analytics and feedback to tutors (Herodotou et al., 2023).

The sixth element of fostering collaborative cultures has been one of the success narratives of AIDA, as KMi and IET (and subsequently Digital Services) worked collaboratively together from the start with the various iterations of AIDA and the evaluations. On an institutional level this element is one of three enablers of the new OU Strategy 2027-2032 as there is a recognition that working at scale and being flexible and agile are not necessarily entwined.

In line with our DBR approach throughout AIDA we developed iterative processes and tested and piloted small scale innovations in each of the six Studies, and used the feedback from both students and staff to refine AIDA. Similarly, building on over a decade of research on learning analytics applications at the OU, there was a strong sense of using data for decision making (Herodotou et al., 2020, 2023; Hlosta et al., 2022; Nguyen et al., 2020; Olney et al., 2021). Building on the work initiated with the creation of OU Analyse in 2011 there are two main steps that we would highlight. Firstly, in 2019 the OU appointed a Chief Data Officer whose office coordinates the use of data for OU business. One of their main activities has been to design and develop an enterprise data hub to eventually house all important OU data. Secondly, a second enabler for the current OU strategy of 'University Priorities' is further improving the data and understanding of our students. In Studies 1-3 no use data for decision-making was available as AIDA was available for participants, but we used interviews and self-report data to focus, understand, define, conceive, and subsequently build the first workable version of AIDA. From Study 4 onwards extensive use of metrics (e.g., time spent on task, ability to complete task), prompts, and interaction data were used to inform how the various functionalities of AIDA were actually used by different (sub)groups of users. Finally, at this point no efforts have been made to build system thinking capacity and institutionalise system thinking.

5. Discussion

Together, these six studies provide a comprehensive system of

Table 4
Most liked AIDA functions for those in the experimental group (1-5).

	% of students used respective functionality	M	SD
The QUIZ function was useful.	88	4.38	.68
The EXPLAIN function was useful.	91	4.33	.66
The CHAT function was useful	85	4.29	1.05
The DYNAMIC TEXT function (simplify or expanding course content) was useful.	88	4.17	.93
The CHALLENGE function was useful.	70	4.09	.73
The REVISE function was useful.	76	4.08	.76

N = 33.

Table 5
Sharples's agenda for a systems approach to AI.

	Key Actions	Study 1	Study 2	Study 3	Study 4	Study 5	Study 6	Institutional
1. Build awareness and leadership support	-Educate leaders in systems thinking -Highlight benefits for managing complexity and sustainable innovation	No	No	Yes (Agreed proposal for development of AIDA with PVCs)	Yes (QEI reports shared with PVCs)	Yes (Demos and various senior level engagement)	Yes (results presented to VCE, Council, shared with PVCs and those responsible for change initiatives)	AIDA Steering Board chaired by PVC-R&I. Setting up of the AI Steering Group sponsored by 3 OU Exec Members New (draft) OU Strategy 2027-2032 specifically builds AI into three core themes Two Senate papers on Responsible AI and use of AI in Assessment.
2. Develop champions	-Identify influential individuals (e.g., leaders, administrators, teachers) -Empower them to advocate for systems approaches	No	No	No	No	Yes (Demos with academic staff in Faculties, embedding AI in teaching (FBL).	Yes (Induction project developed with WELS staff. Collaboration).	Setting up of an OU GenAI Community of Practice allowing for the sharing of knowledge and for champions to emerge. Specific roles in AI Steering Group for community engagement Development of 'AI Champions' across the OU.
3. Map the institution as a system	-Use tools like system maps, causal loop diagrams -Understand interconnections across functions -Involve diverse stakeholders	No	No	No	No	No	Yes	AI Steering Group structure represents high-level description of the institution.
4. Define innovation goals	-Set clear, shared objectives for innovation -Align innovation goals with institutional mission and values	No	Yes (4 main services)	Yes (Created Design Specifications and principles)	Yes	Yes	Yes	The AI Steering Group has a process for receiving suggestions and prioritising/ advising on these. OU Strategy: "Lead in the responsible use of AI in education, ensuring that it strengthens learning, teaching, research and student support while protecting academic integrity, inclusion and human judgement" New OU Strategy includes specific KPIs on student success, learning innovation, and AI.
5. Identify leverage points	-Locate small changes with high potential impact using system maps -Prioritise initiatives by feasibility and impact	No	No	No	Yes (OLC as first context for AIDA, involve Digital Services)	Yes	Yes	The OU has over the last five years been conducting quick pilots with AI to ascertain potential wins. Workshops organised as part of OU Strategy identify Innovation Pipeline issues and how to support staff and students with critical AI literacy
6. Foster collaborative cultures	-Break down silos -Form interdisciplinary teams -Enable open dialogue and stakeholder engagement	Yes	Yes	Yes	Yes (work with staff to alpha-test)	Yes (work with students- on beta test)	Yes	The new OU strategy 'University Priorities' has 'People and Culture' as one of three enablers. Removing silos is a key goal within this strand.

(continued on next page)

Table 5 (continued)

	Key Actions	Study 1	Study 2	Study 3	Study 4	Study 5	Study 6	Institutional
7. Develop iterative processes	-Pilot small-scale innovations -Use feedback loops for refinement and scaling	No	Yes	Yes (first design iteration)	Yes	Yes	Yes	The precursor to the work described here (2020-2023) was 'Test and Learn'. A multi-million pound activity where short projects could be quickly setup and run with the aid of outside consultants.
8. Build systems thinking capacity	-Provide training and professional development -Partner with external systems-thinking experts	No	No	No	No	No	No	
9. Use data for decision-making	-Monitor KPIs and systemic health metrics -Analyse trends and feedback loops	No	No	No	Yes (time spent on task, able to complete task)	Yes (same)	Yes (OpenLearn statistics)	Appointment of Chief Data Officer in 2019. OU strategy of 'University Priorities' is further improving the data and understanding of our students.
10. Institutionalise systems thinking	-Embed principles in strategy, policy, and operations -Include systems thinking in curricula for staff and students -Commit to ongoing evaluation and learning	No	No	No	No	No	Yes (Recommendations for ongoing evaluation and monitoring picked up by Digital Services?)	

learning account of how a large distance learning institution like the Open University (OU) has designed and evaluated their own institutional GenAI assistant, with substantial lessons learnt through this. Working reactively and proactively, this research has identified that distance learning institutions and their students have unique requirements and potential benefits to be gained from building solutions tailored to these needs (Herodotou et al., 2020, 2023; Hlosta et al., 2022; Olney et al., 2021). Key contributions include a multi-phase co-design methodology of six studies, empirical evidence that provides initial explorative support for the educational value of contextually tuned AI interactions, and reflections on engaging senior management and the wider workforce with the new system and findings. As new applications of GenAI such as AIDA develop and are shared, the findings contribute to ongoing debates about the future of learning systems, offering practical pathways for (distance learning) institutions seeking to adopt GenAI ethically, transparently, and at scale.

In terms of RQ1, across the six Studies initial perceptions of AIDA were generally positive, with the majority of students recognising its potential for 24/7, context-specific support embedded within the OU learning environment. This aligns with prior work indicating that immediacy, contextual relevance, and personalisation are valued features in AIDA (Rienties et al., 2024). However, early-stage perceptions in Studies 1-3 were shaped largely by conceptual exposure rather than actual hands-on use of AIDA, and included concerns over academic integrity, ethical risks, and data privacy (Rienties et al., 2025; Rienties et al., 2025), whereby around 20% of students were highly sceptical whether the OU should develop and use AIDA.

Consistent with Diffusion of Innovations Theory (Jin et al., 2025), some of these early reservations diminished following actual use in Study 5 (Rienties, Ullmann, et al., 2025), and in particular in our explorative Study 6 across two OLC courses. Indeed in Study 6 participants who actually used AIDA indicated mostly positive attitudes post-interaction. The hands-on use of AIDA's quiz, explain, and chat functions appeared to shift perceptions from abstract potential to

tangible usefulness, highlighting the importance of experiential engagement in fostering acceptance of AI tools in distance education.

In terms of RQ2, Study 6 provided a first opportunity to observe AIDA's impact "in the wild," without facilitator guidance. Notably, 96% of students irrespective of control or experimental condition expressed interest in having AIDA available in their formal studies, reinforcing the perceived educational value of AI assistants when integrated into authentic learning settings. Although participants supported with AIDA had twice longer usage of OLC relative to those who did not have AIDA, no significant differences were found on overall time and various other learning process data. One possible reason for the lack of significant difference in process and outcome data could be the explorative task design, whereby students were asked to explore the OLC course at their own leisure. Another reason might be that the sandbox environment was different from the main teaching and learning environment that our distance learning students use regularly. No credits or external financial reward were given, and hence most learners just briefly looked at the OLC course.

Mapping the six Studies of AIDA against Sharples (2025) embedded systems framework in Table 5 revealed that sustained progress depended on a combination of bottom-up co-design and top-down strategic alignment. Key enabling factors included senior executive sponsorship, cross-unit collaboration between the two R&D units of KMi and IET, Digital Services, and Faculties, and the iterative DBR cycles to refine the functionality of AIDA in response to user feedback from both students and educators. These institutional conditions mirror lessons from prior large-scale innovation adoption in learning analytics (Herodotou et al., 2020, 2023; Olney et al., 2021), underscoring that technical capability must be coupled with organisational readiness and appropriate governance structures. At the same time, the analysis identified gaps—particularly in building systems-thinking capacity and institutionalising systemic change—suggesting that early AIDA work operated in a "pilot project" mode. As institutional awareness and evidence of impact have now been established, future work should focus on

embedding GenAI initiatives into organisational processes, scaling beyond pilot contexts, ensuring continuous ethical oversight in line with the OU's Responsible AI Framework, and ongoing consideration of AI-related concerns such as poor AI outputs quality and impact on staff workload (Boroowa & Herodotou, 2022).

At the time of writing (April 2026) further developments of AIDA for students have unfortunately stalled. The induction course pilot with AIDA in September-October 2025 was used by 354 students in one Faculty. Survey findings showed that AIDA supported early-stage orientation during induction, with the most common queries being related to navigating university's systems, understanding the course structure, and clarifying study expectations. No further testing with new cohorts of students was agreed beyond this trial.

As with many large organisations, the transition from innovation to institution-wide implementation is rarely straightforward. The authors continue to collaborate closely with relevant production teams as efforts progress to integrate the AI capabilities outlined in this paper within OU's Moodle platform. Alongside technical considerations, organisational priorities, resource availability, and implementation capacity all shape the pace and scope of deployment. More broadly, the process of scaling AIDA has highlighted the need to balance ambitions for rich, pedagogically informed AI support with the practical requirements of delivering robust, sustainable, and lower-risk solutions at scale.

While substantial awareness of the (potential) impact of AI on teaching and learning is recognised at a senior level at the OU, as evidenced by the new (draft) OU strategy that has AI embedded throughout the core provision of teaching and learning and two recent Senate papers on responsible GenAI, the innovation pipeline to introduce GenAI to students at scale at present is progressing slower than the authors would wish. This is not the first time that this has happened (Herodotou et al., 2020, 2023; Sharples, 2023; Sharples et al., 2015), and this will not be the last time either in complex and large distance organisations like the OU, with many stakeholders and a complex digital technology infrastructure. In many organisations the last-mile delivery of promising innovations that need to be scaled and become business as usual (Dawson et al., 2018; Vigentini et al., 2020) often struggle to make it all the way to the end-user who could potentially benefit most from personalised AI

support in teaching and learning. Therefore, in the remainder of this discussion the four authors will reflect on the lessons learned from nearly two years of developing, testing, evaluating AIDA with students and staff.

5.1. Lessons learned

[Lesson 1] Bringing together a diverse group of experts and facilitating their collaboration is crucial for developing an effective institutional strategy in response to AI, but not sufficient to embed AI into business as usual.

As evidenced in these six Studies and elsewhere implementing GenAI appropriately in teaching and learning is inherently complex, especially across a large institution. As outlined in Fig. 3 the AI Steering Group aggregates input from across the university and provides recommendations and advice to the OU executive on strategy, compliance, engagement, use cases and technical foundations and enablers. Staff come from: the OU innovation units, Digital Services (infrastructure, enterprise data, VLE, Information Security), Information Rights, Data Protection, Organisational Training, Marketing and Communications, HR and the OU faculties. One distinct role within the Steering Group is to coordinate OU policy related to teaching and AI.

We have also setup three measures to engage broadly with staff across the university. Firstly, a Generative AI Community of Practice (GenAI CoP) was setup, by one of the paper authors, in July 2023 to enable the sharing of activities by any OU staff related to the application of AI to OU teaching. The GenAI CoP now has over 650 members with over 80 attending the monthly meetings. Recent meetings have covered AI literacy for students, enhancing AI capabilities and applying AI in assessment grading.

Secondly, dedicated intranet pages have been setup which describes a scenario of GenAI in use accompanied by screencasts and examples. Finally, dedicated awareness raising sessions on GenAI in general have been conducted over two + years with units across the OU including finance, HR, and a number of schools and Faculty leadership teams.

[Lesson 2] Senior institutional oversight on AI innovation activities has two benefits: stronger alignment of innovation/piloting/evaluation activity to institutional priorities and greater awareness of the activity by the institution's

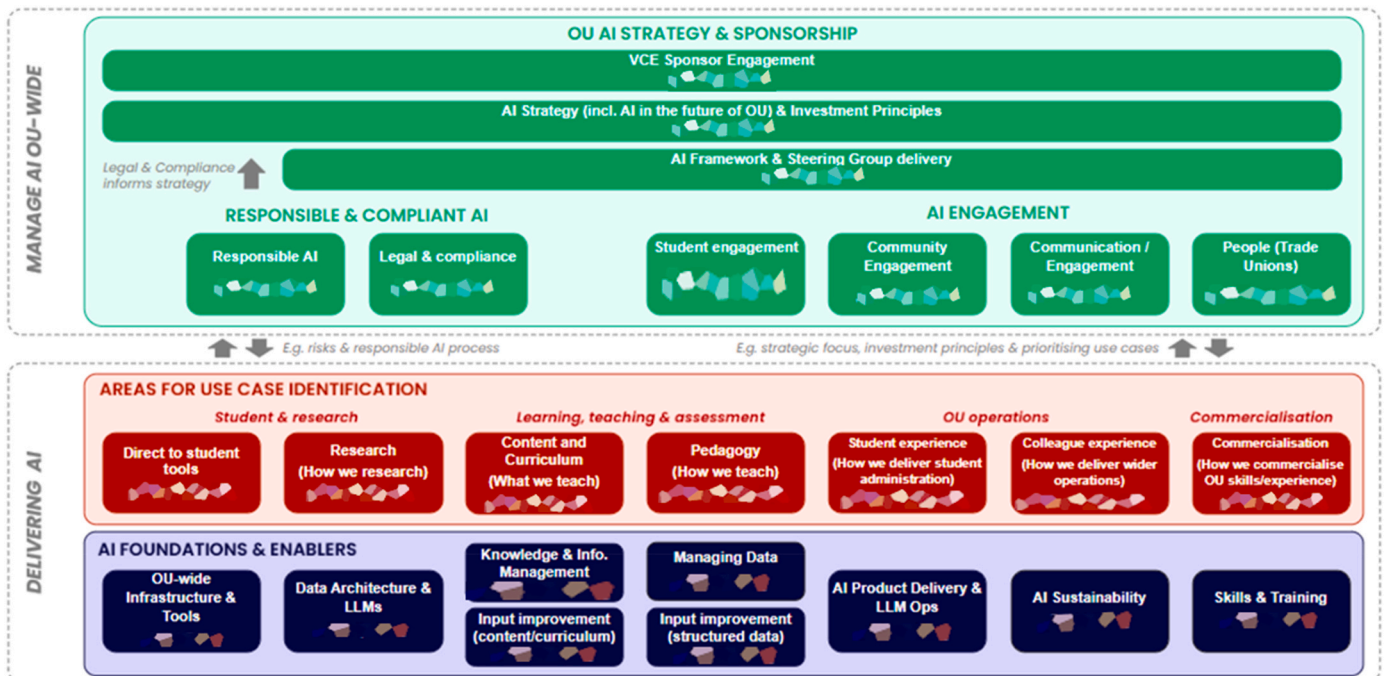


Fig. 3. An overview of the main roles and activities in OU AI Steering Group

Source: <https://openuniv.sharepoint.com/sites/data-info-gov/data-hub/SitePages/OU-AI-Framework-and-Steering-Group.aspx>.

executive.

[Lesson 3] *It is essential to ensure broad institutional buy-in on the potential of AI, the associated challenges and the sharing of lessons learnt to enable take-up and use for organisational and student benefit.*

The new (draft) OU Strategy 2027-2032 specifically builds AI into three core themes of creating social impact, supporting student success, and reaching new communities through innovation and partnership. Specifically, the new Strategy proposes to “lead in the responsible use of AI in education, ensuring that it strengthens learning, teaching, research and student support while protecting academic integrity, inclusion and human judgement”. Indeed two recent senate papers on Responsible AI and use of AI in Assessment in March 2026 indicate that across the institution and amongst senior management there is a strategic push to embed AI into teaching and learning.

[Lesson 4] *It is essential that universities have dedicated R&D units that have the capability and capacity to explore new pedagogies and technologies outside of mainstream teaching and learning operations.*

To achieve this, universities need to have in place services that work for this purpose; that is dedicated staff time to research and develop new methods, tools and solutions that can tackle key university challenges and ensure broader tech innovations are leveraged for the benefit of teaching and learning. In this paper, we showcased that the OU had such capacity in place through academics and researchers from two units with expertise in pedagogy, evaluation and tech development (IET and KMi). Developing and executing strategy to deal with the paradigm shifting nature of Generative AI will require that expertise from all these areas - pedagogy, AI, scalable IT infrastructure and services, educational content development and management – plus finance and human resources.

5.2. Limitations and future research

While these six studies offer a broad and integrated perspective, some limitations should be recognised. The DBR work was carried out within a single, large-scale distance learning university in the UK, which may restrict how far the findings can be applied to other settings. Differences in institutional regulations, digital infrastructure, and learner profiles across educational contexts could influence both the practicality and the effectiveness of AIDA. In addition, the actual deployment of AIDA has yet to be tested with substantially larger groups of students in actual qualifications and modules, beyond the two free none-credit bearing OLC courses of Study 6 and the optional induction course pilot.

The two-year + research window, though relatively long, may still be insufficient to observe enduring adoption patterns or to account for the rapid evolution of generative AI tools (Bond et al., 2024; Giannakos et al., 2025). Given the fast pace of developments in AI—particularly in natural language processing and adaptive learning—it is likely that future versions of AIDA-like features in the VLE will differ considerably from the one examined in this study. Furthermore, testing “real” AIDA applications with credit-bearing modules with “real” distance learning students in authentic contexts might provide further evidence of the affordances and limitation of GenAI support for distance learners.

Third, the reflections of how the OU has implemented GenAI and AIDA in particular was using Sharples (2025) model were solely based upon the reflections of the four authors. This is an inherent limitation as others might have different interpretations of the six Studies and the embedded system approach at the OU.

5.3. Future Research

Our six Study approach points to several avenues for further investigation. One priority would be a longitudinal study to track the ongoing influence of AIDA-like features in the VLE on student engagement and achievement within their actual modules and overall qualifications. Such research could clarify whether learners sustain their use of AI-based tools over time and how these tools contribute (if any) to their

academic performance, offering insights into their long-term value and potential limitations. Another useful direction would be to replicate this research in a variety of institutions, including those operating under different educational models (e.g., campus-based universities, professional training providers). Cross-institutional comparisons could shed light on the contextual factors that shape both the effectiveness of AI and its reception by a range of stakeholders, including students, staff, and the broader economic and social environment.

Open data and ethics statement

The six Studies were approved by the OU HREC (2023-0183-4, 2024-0660-2, 2025-0831-3). Informed consent was obtained from all participants prior to data collection, and all procedures complied with relevant institutional and UK ethical guidelines. Participants' privacy and confidentiality were strictly maintained, and all data were anonymised prior to analysis.

Due to the sensitive nature of the data and ethical considerations related to participant privacy, the datasets of the six studies are not publicly available. However, anonymised data may be made available from the corresponding author upon reasonable request, subject to institutional and ethical approvals.

CRedit authorship contribution statement

Bart Rienties: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tim Coughlan:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **John Domingue:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Conceptualization. **Christothea Herodotou:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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